How-To Library

Informatica

Creating a Dashboard for Monitoring Enterprise Data Preparation User Activities

© Copyright Informatica LLC 2018, 2021. Informatica and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html

Abstract

You can use the Apache Zeppelin notebook application to create a dashboard for monitoring Enterprise Data Preparation user activities. Review this article to learn how to create a monitoring dashboard.

Supported Versions

- Enterprise Data Preparation 10.2.1, 10.4.x

Table of Contents

Overview. 2
Configure the Enterprise Data Preparation Service to Log User Activity Data. 3
Create a JDBC Interpreter in Zeppelin. 3
Create a Notebook in Zeppelin. 6
Explore the User Activity Data and Create Queries. 8
Query Examples. 9

Overview

You can use Apache Zeppelin to create a dashboard to monitor the activities performed by Enterprise Data Preparation users.

Zeppelin is a web-based notebook application that is integrated with Enterprise Data Preparation. You might use Zeppelin to visualize published assets in Enterprise Data Preparation.

When you enable user activity logging in Enterprise Data Preparation, the application logs every action each user completes in the Model repository associated with Enterprise Data Preparation. Using Zeppelin, you can visualize user activity data using tables, charts, and graphs.

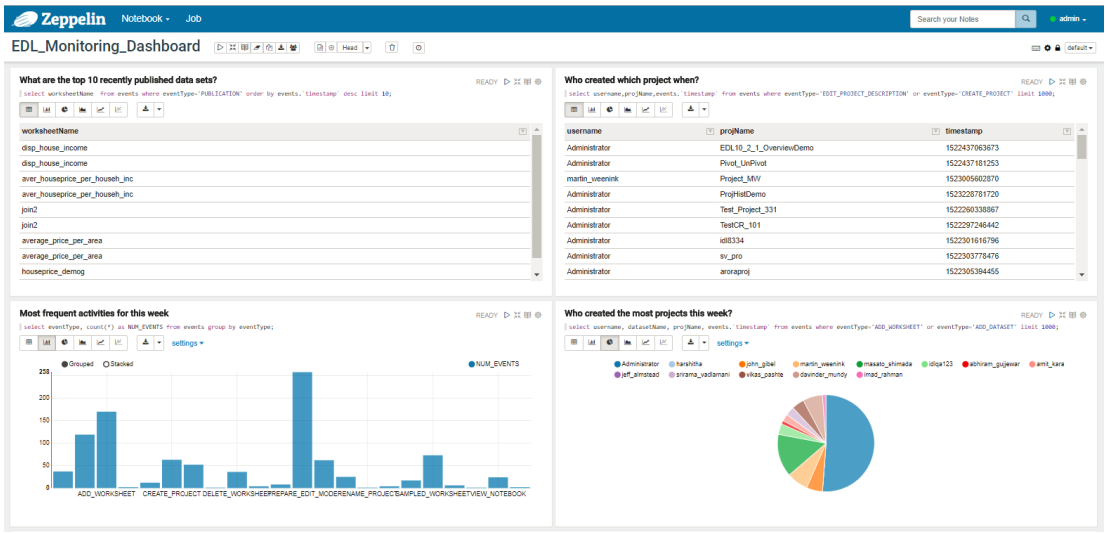
You might want to create a dashboard to view the following Enterprise Data Preparation usage statistics:

- Most active users.
- Most heavily used projects.
- Import, export, and publish activity details for each user.
- Volume of data assets created by each user.
- Activity trends over a specific period of time.

You create the dashboard as a notebook in Zeppelin. You create SQL queries in the notebook to retrieve user activity data stored in the Model repository.

To run the queries, you create a JDBC interpreter in Zeppelin, which enables Zeppelin to use the JDBC driver provided with Enterprise Data Preparation to retrieve log data from the Model repository.

The following image shows a notebook created as an Enterprise Data Preparation monitoring dashboard:



For more information about using Zeppelin and creating notebooks, see the Zeppelin documentation at the following link:

https://zeppelin.apache.org/

Configure the Enterprise Data Preparation Service to Log User Activity Data

Use the Administrator tool to configure the Enterprise Data Preparation Service to log user activity data.

1. In the Administrator tool header, click the **Manage** tab > **Services and Nodes** view.
2. Select the Enterprise Data Preparation Service.
3. In the Event Logging section in the Properties tab, enter the following user event logging properties:

Property	Description
Log User Activity Events	Set to true to enable user event logging.
JDBC Port	JDBC port to use to retrieve user activity event data. Enter an available port on the node where the Enterprise Data Preparation Service runs.

4. Recycle the Enterprise Data Preparation Service.

Create a JDBC Interpreter in Zeppelin

Create a JDBC interpreter in Zeppelin. The interpreter enables Zeppelin to use the JDBC driver provided with Enterprise Data Preparation to run SQL queries to retrieve log data from the Model repository.

For more information about creating interpreters, see the Zeppelin documentation at the following link:

https://zeppelin.apache.org/docs/0.7.3/manual/interpreters.html

1. Log in to the Zeppelin user interface.
2. Select **Interpreter** from the user name menu.

3. Click **Create**.
4. Enter a descriptive name for the interpreter.
5. Select **jdbc** from the Interpreter Group menu.
6. Click **edit**.
7. Edit the required connection properties in the **Properties** section of the page.

   The following table describes the properties you set:

Property	Value	Description
default.driver	`com.informatica.idl.jdbc.JDBCDriver`	The Enterprise Data Preparation JDBC driver. The driver is bundled with Enterprise Data Preparation.
default.user	*<user name>/<security domain>*	The Enterprise Data Preparation administrator user name and the security domain to which the user belongs.
default.password		The administrator user password.
default.url	`jdbc:edl://`*<host name>*`:`*<JDBC port>*	The URL for the Enterprise Data Preparation JDBC driver. *<host name>* is the host name of the node where the Enterprise Data Preparation Service runs. *<JDBC port>* is the value set in the JDBC Port property under Event Logging Options for the Enterprise Data Preparation Service.

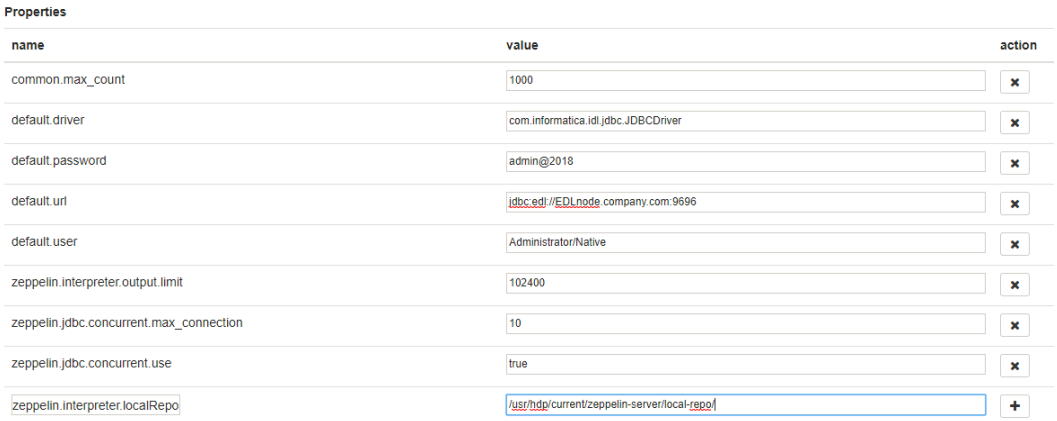
8. Add the `zeppelin.interpreter.localRepo` property to the **Properties** section of the page.

   The following table describes the property configuration:

Property	Description
zeppelin.interpreter.localRepo	Set to any local directory in the Zeppelin host machine. Zeppelin creates a subdirectory within the directory to which to copy dependent artifacts.

9. Click the **+** icon in the Action column after you add the `zeppelin.interpreter.localRepo` details.

The following image shows the edited properties in the **Properties** section, including the `zeppelin.interpreter.localRepo` property:



Properties

name	value	action
common.max_count	1000	×
default.driver	com.informatica.idl.jdbc.JDBCDriver	×
default.password	admin@2018	×
default.url	jdbc:edl://EDLnode.company.com:9696	×
default.user	Administrator/Native	×
zeppelin.interpreter.output.limit	102400	×
zeppelin.jdbc.concurrent.max_connection	10	×
zeppelin.jdbc.concurrent.use	true	×
zeppelin.interpreter.localRepo	/usr/hdp/current/zeppelin-server/local-repo/	+

10. Browse to the following directory on the Enterprise Data Preparation host machine:

    `<Enterprise Data Preparation installation directory>/services/IDLService/Binaries/solr-7.1.0/jdbc`

11. Copy the following .jar files containing the JDBC driver and dependent artifacts from the Enterprise Data Preparation host machine to a directory on the Zeppelin host machine:
    - com.infa.products.idlservice.solr.plugins.jdbc-[10.x]-SNAPSHOT.jar, where [10.x] represents the complete Informatica build number
    - commons-io-2.4.jar
    - commons-math3-3.6.1.jar
    - httpclient-4.5.3.jar
    - httpcore-4.4.6.jar
    - httpmime-4.5.3.jar
    - jcl-over-slf4j-1.7.7.jar
    - noggit-0.8.jar
    - slf4j-api-1.7.7.jar
    - solr-solrj-7.1.0.jar
    - stax2-api-3.1.4.jar
    - woodstox-core-asl-4.4.1.jar
    - zookeeper-3.4.5.jar
12. Add the relative path to each .jar file on the Zeppelin host machine to the **Dependencies** section of the page.

    Click the **+** icon to add the path to each .jar file.

The following image shows the relative path to each .jar file added to the **Dependencies** section of the page:

Dependencies

artifact
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/commons-io-2.4.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/commons-math3-3.6.1.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/httpclient-4.5.3.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/httpcore-4.4.6.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/httpmime-4.5.3.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/com.infa.products.idlservice.solr.plugins.jdbc-10.2.100.620.376-SNAPSHOT.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/jcl-over-slf4j-1.7.7.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/noggit-0.8.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/slf4j-api-1.7.7.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/solr-solrj-7.1.0.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/stax2-api-3.1.4.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/woodstox-core-asl-4.4.1.jar
/export/home/idlui/dkhan/zeppelin-0.7.2-bin-all/local-repo/solr_jdbc_idl/zookeeper-3.4.5.jar

13. Click **Save** to save the interpreter.

Create a Notebook in Zeppelin

Create the monitoring dashboard as a notebook in Zeppelin.

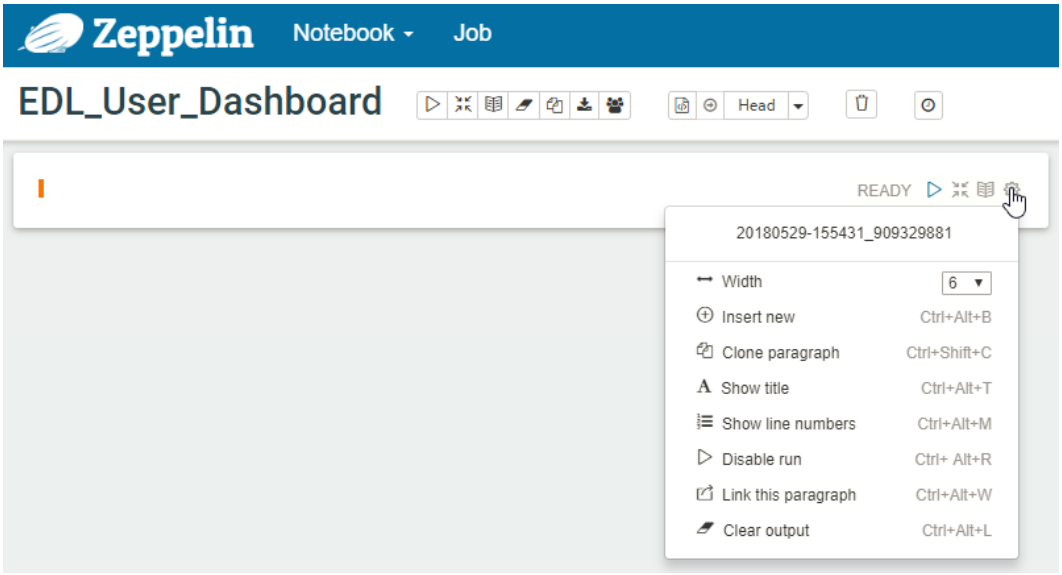
A Zeppelin notebook consists of one or more paragraphs. You can think of a paragraph as a panel within the dashboard. You add a SQL query to each paragraph to retrieve the user activity data to display in the paragraph.

Use the following instructions to create a create a paragraph within a notebook.

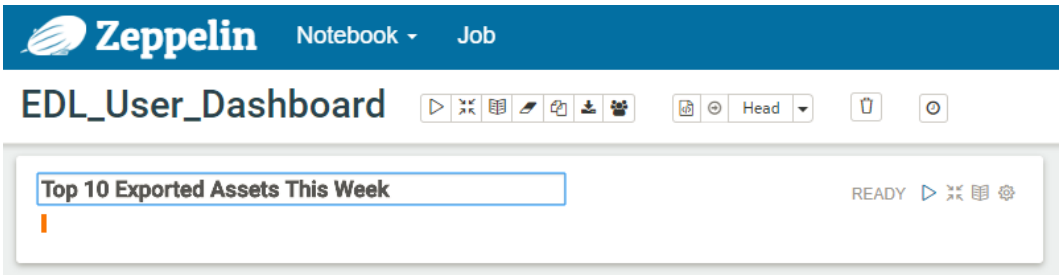
1. Log in to the Zeppelin user interface.
2. Select **Create new note** from the **Notebook** menu.
3. Enter a unique name for the monitoring dashboard notebook.
4. Select your JDBC interpreter from the Default interpreter menu.
5. Click **Create Note**.

   Zeppelin creates the note with an empty paragraph.

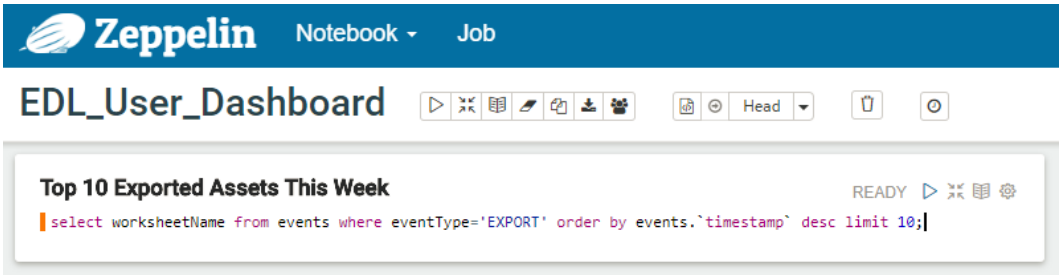
6. Select **Width** from the gear icon within the paragraph, and then set the paragraph width, as shown in the following image:



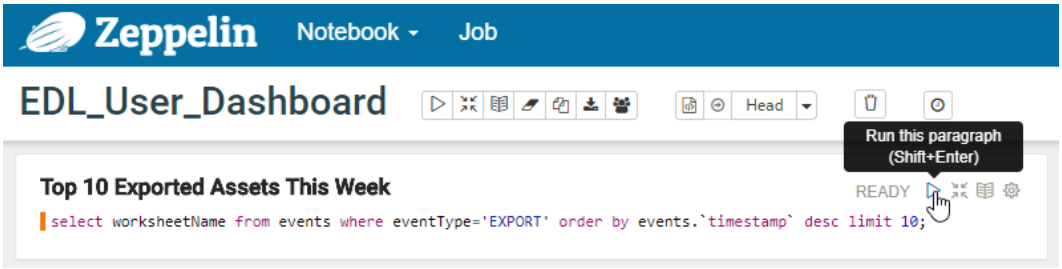
7. Select **Show Title** from the gear icon within the paragraph, and then enter a title for the paragraph, as shown in the following image:



8. Type the SQL query next to the red bar below the title, as shown in the following image:



9. Click the arrow in the paragraph to run the query, as shown in the following image:



Explore the User Activity Data and Create Queries

Enterprise Data Preparation writes user activity data to the `events` table in the Model repository database. To create SQL queries to retrieve data from the `events` table, you must first explore the data stored in the table.

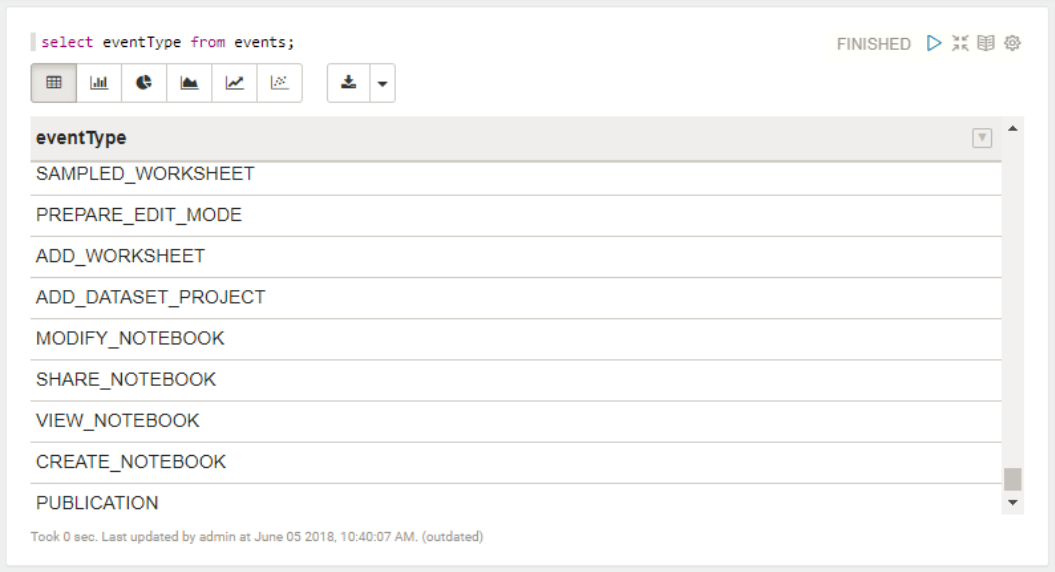
Once you understand the structure of the data you want to retrieve, you can create SQL queries to use in the paragraphs in your monitoring dashboard notebook.

To explore the data in the `events` table, create a notebook in Zeppelin, and then run a query to retrieve the user activity event types from the table. You then run another query to retrieve the columns for a specific event type from the table.

1. Log in to the Zeppelin user interface, and then create a new notebook.
2. In the empty paragraph in the notebook, run the following query to retrieve the list of all user activity event types:

   ```
   select eventType from events;
   ```

   The following image shows the query results in a notebook paragraph:

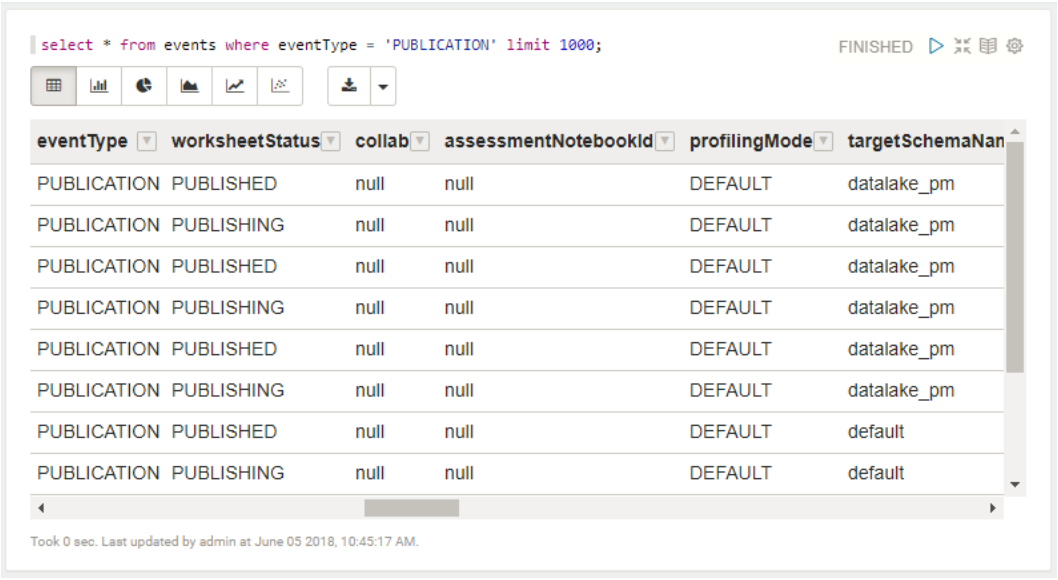


3. After you retrieve the list of event types, run the following query to retrieve the columns for a specific event type:

   ```
   select * from events where eventType = <eventType> limit <value>;
   ```

   You must specify a numeric value for `limit` to enable the query to retrieve the columns for a specific event type. Specify a large number to ensure that you retrieve all columns.

   The following image shows the results for a query to retrieve the columns for the `PUBLICATION` event type, with the value for `limit` set to `1000`:



Query Examples

The following table contains sample queries you can use in paragraphs to retrieve user activity data:

Data to Retrieve	Query
What are the top 10 recently published data sets?	`select worksheetName from events where eventType = 'PUBLICATION' order by events.`timestamp` desc limit 10;`
Who created which project, and when?	`select username,projName,events.`timestamp` from events where eventType = 'EDIT_PROJECT_DESCRIPTION' or eventType = 'CREATE_PROJECT' limit 1000;`
Who added which asset to which project, and when?	`select username, datasetName, projName, events.`timestamp` from events where eventType = 'ADD_WORKSHEET' or eventType = 'ADD_DATASET' limit 1000;`

Authors

Swathi N.R.

Dan Hynes